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CFI - Care: Collaborative Flow of Intelligence for Care

Graduation Project Thesis

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Abstract

Healthcare today is characterized by fragmented experiences where patient medical information is often dispersed across various institutions and providers. This project proposes a comprehensive, patient-centric healthcare application designed to consolidate all medical reports into a single, secure, and unified platform. The system employs Role-Based Access Control (RBAC) to ensure privacy and controlled information sharing. To enhance clinical decision-making, the platform features a medical history visualization graph for tracking health trends and integrates a Retrieval-Augmented Generation (RAG)-enabled AI model to generate diagnostic reports and insights. By transforming isolated data into an interconnected ecosystem, this project aims to establish meaningful relationships between patients, caregivers, and doctors for personalized healthcare delivery.

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Chapter 1

Introduction

Today, patients often face fragmented healthcare experiences, with their medical information dispersed across multiple institutions and providers . This project proposes a patient-centric healthcare application designed to consolidate all medical reports within a single, secure, and unified platform . The system empowers patients to manage their

health through continuous data collection, secure communication, and controlled information sharing using Role-Based Access Control (RBAC) to ensure privacy, consent, and data integrity . Additionally, by integrating features such as medical history visualization and Retrieval-Augmented Generation (RAG)-enabled AI, the proposed system transforms previously scattered and isolated medical data into a unified, interconnected ecosystem . This integration establishes meaningful relationships between patients, healthcare providers, and medical records, promoting continuous, coordinated, and personalized healthcare delivery .

Purpose

The primary purpose of this project is to develop a unified healthcare platform that securely consolidates patients' medical records from multiple institutions . By leveraging technologies such as RAG-enabled AI and graphical data visualization, the project aims to enhance clinical insight and support medical decision-making . Furthermore, the platform seeks to empower patients to monitor their own health data while maintaining full control over information sharing through robust RBAC mechanisms . Ultimately, this initiative strives to ensure privacy, data integrity, and interoperability across the healthcare ecosystem to enhance patient trust and system reliability .

Objective

The key objectives of this project are as follows:

- To develop a unified healthcare platform that securely consolidates patients' medical records from multiple institutions .
- To empower patients to manage and monitor their own health data while maintaining full control over information sharing through RBAC mechanisms.
- To enable secure and efficient communication between patients, doctors, and caregivers .
- To provide data visualization tools, including a medical history graph, to help healthcare providers interpret patient data trends and clinical progress .
- To integrate RAG-enabled AI capabilities to generate automated reports and actionable insights that support doctors in diagnosis and medical decision-making.
- To ensure privacy, data integrity, and interoperability across the healthcare ecosystem to enhance patient trust and system reliability .

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References

- [1] *FHIR HL7 Documentation*, <https://www.hl7.org/fhir/>, Accessed: 2025-04-04.